



# SOTERIA

## From Chaos to Clarity Disaster Response at the Speed of AI

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## Problem Briefing

# Information Bottlenecks Cost Lives

*"In a disaster, response teams are flooded with unstructured, chaotic information. Every minute spent making sense of this data costs lives."*

— SOTERIA RESPONSE RESEARCH



## Critical Pain Points

# Legacy Response Cognitive Overload

- 1. **Unstructured Floods**  
Victims send frantic voice notes in local dialects and blurry photos rather than structured data.
- 2. **Manual Sifting**  
Commanders manually sift through hundreds of tweets, calls, and field reports under pressure.
- 3. **Reactive Dispatch**  
Prioritization becomes reactive: the loudest request gets attention instead of the most urgent.
- 4. **Offline Blindness**  
Volunteers arrive on-site without knowing specific hazards as cellular networks fail when needed most.



### Field Data Flood

Frantic messaging, disconnected platforms, and blurry photos overwhelm responders, delaying critical dispatch.



# Existing Tools Require Manual Structuring That Doesn't Scale

| Category          | Current Solutions                                                         |
|-------------------|---------------------------------------------------------------------------|
| MASS INTAKE       | Government helplines are manual and language-limited                      |
| SOCIAL MONITORING | Social media monitoring is buried in noise                                |
| SOS APPLICATIONS  | Basic SOS apps require stable internet and lack prioritization            |
| GIS MAPPING       | Static GIS maps show past data                                            |
| VOLUNTEER COORD.  | Volunteer coordination via WhatsApp lacks task matching and live tracking |

Soteria Platform

✓

 Automated multimodal intake

✓

 Uses GenAI to extract structured data

✓

 Offline-first with AI triage scoring

✓

 Real-time predictive heatmaps

✓

 Structured dispatch with AI-verified closure



SOTERIA

#### THE CORE THESIS

## *"The Missing Layer is Automated Understanding, Not Just Data Collection"*

The true bottleneck in disaster response isn't a lack of data. The failure point is the inability to convert chaotic, multimodal human communication into structured, prioritized action fast enough to matter.

We are not building just another SOS application. Instead, we are building the critical missing understanding layer that uses AI to process information instantly, allowing human responders to focus entirely on taking action.

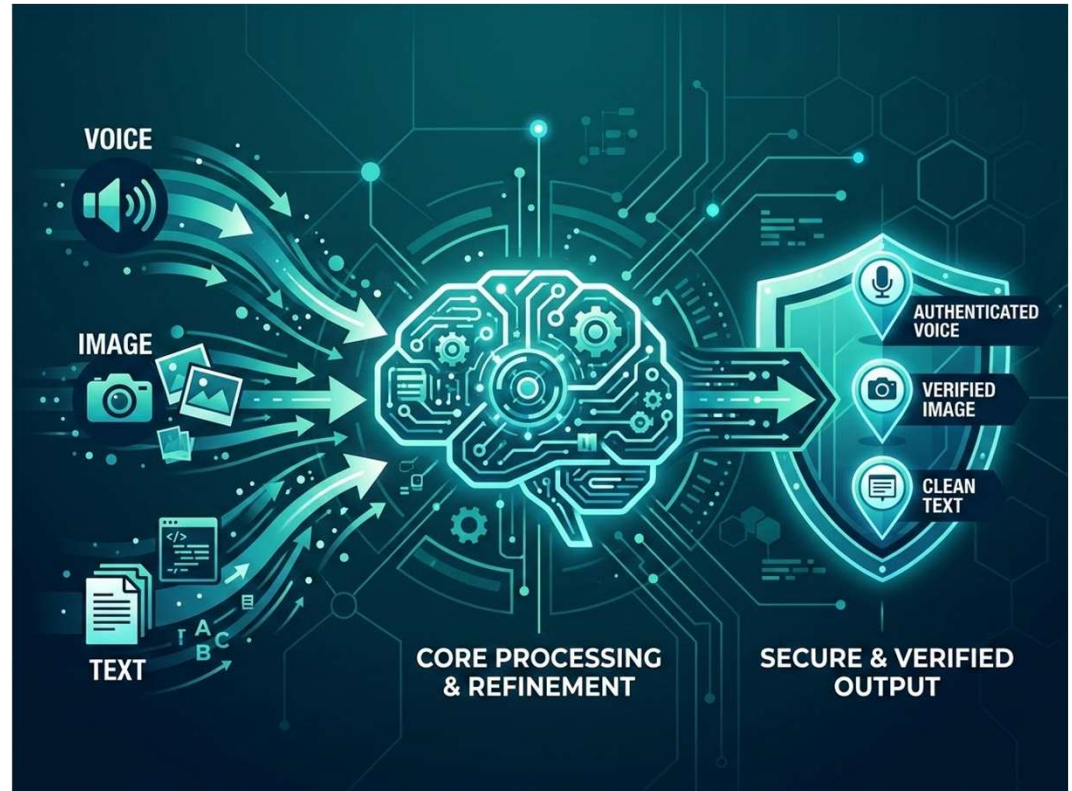


## SOTERIA PLATFORM

# A Unified Intelligence Layer for Real-Time Disaster Triage

Soteria is a unified intelligence layer that ingests raw, unstructured disaster data—including voice, images, social media, and field reports—and converts it into actionable intelligence in real time.

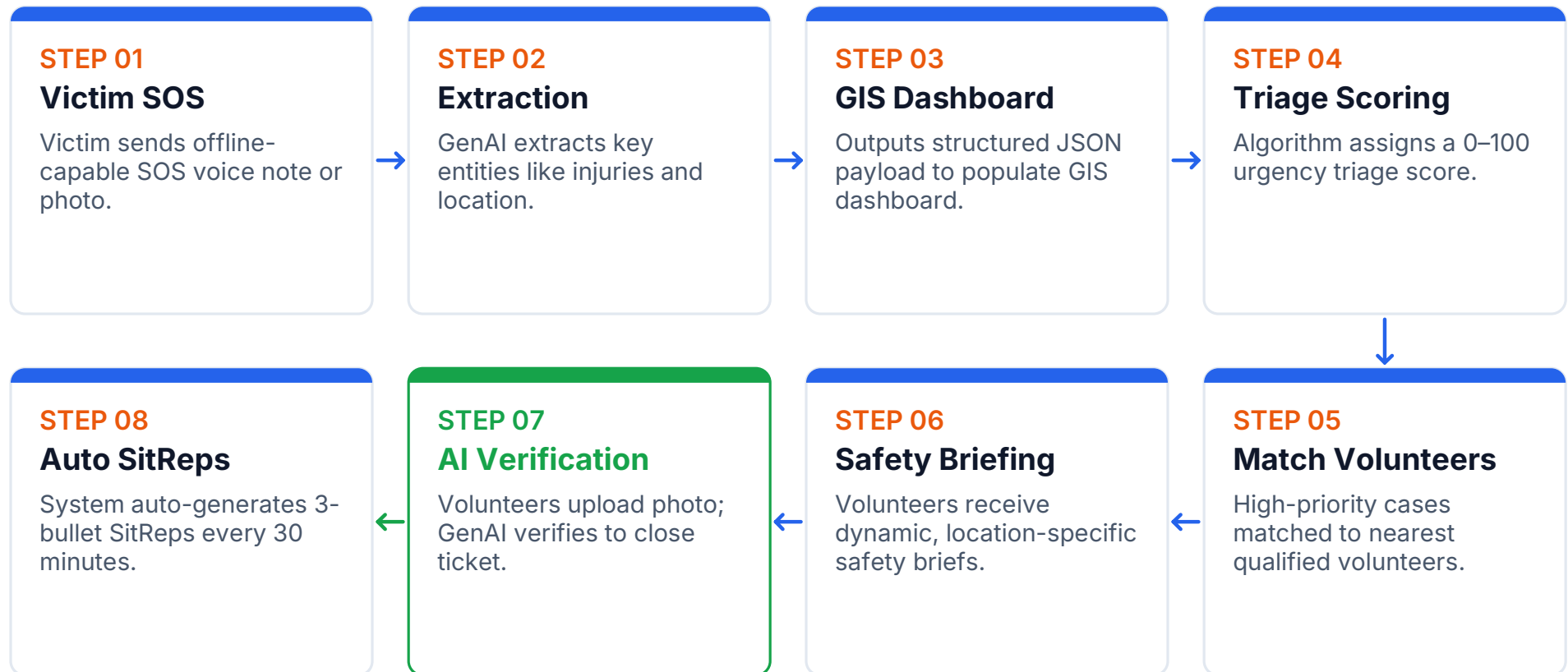
- **01INPUT** — Voice, Photo, Social
- **02EXTRACTION** — GenAI Multimodal
- **03INTELLIGENCE** — Triage & Risk Mapping
- **04ACTION** — Volunteer Auto-Dispatch
- **05OUTCOME** — Prioritized Rescue





# Automated End-to-End Workflow

From SOS to Verified Rescue



# Platform Capabilities

## Core MVP features and advanced future roadmap

### CORE MVP PLATFORM

#### 01 / TRIAGE

##### Multimodal SOS Triage

Converts voice and image inputs directly to structured JSON via GenAI models.

#### 02 / DISPATCH

##### GIS Command Dashboard

Features a hexagonal risk-density heatmap, victim prioritization with a 0–100 urgency score, and volunteer dispatch matching.

#### 03 / ACCESSIBILITY

##### Offline-Capable Citizens PWA

Provides a lightweight, resilient Progressive Web App designed specifically for citizens in low-connectivity disaster areas.



### ADVANCED FUTURE ROADMAP

#### 04 / PREDICTIVE AI

##### Predictive Flood Modeling

Introduces real-time predictive flood propagation modeling using topographical and geospatial elevation data.

#### 05 / RESILIENCE

##### Bluetooth Mesh Relay

Establishes mesh networking infrastructure to guarantee zero-connectivity relays between active devices.

#### 06 / ROADMAP

##### Intelligent Command Tools

Enables dynamic SOP generation, automated 30-minute SitReps for commanders, and AI-verified proof-of-action photo closure.



# The Only Offline-First, Multimodal Triage Engine

Converts raw human distress signals into structured, dispatch-ready intelligence in real time.

## ✓ Multimodal-First

Victims can speak or send photos instead of filling out complex forms under extreme duress.

## ✓ Language-Agnostic

Utilizes advanced real-time AI translation to instantly remove critical dialect and language barriers.

## ✓ Offline-First

Engineered specifically for dead tower zones to maintain critical triaging functions without internet.

## ✓ Predictive Modeling

Uses proactive risk modeling to anticipate cascading hazards rather than merely reacting post-event.

## ✓ Closed-Loop Dispatch

Features secure end-to-end routing with verified closure including photo-based rescue receipts.



Target User Ecosystem

# Designed for Citizens, Volunteers & Commanders

## 01 / CITIZENS IN DISASTER ZONES

### Mitigating Offline Isolation

Enables trapped victims to record offline voice notes. The system automatically caches, syncs, and triages these critical messages the moment network signal returns.

## 02 / VOLUNTEERS & NGOS

### Hazard Exposure & Deployment

Ground teams receive dynamic safety SOP briefs for unfamiliar hazards, while NGOs utilize a live GIS heatmap to allocate volunteers to high-density risk zones.

## 03 / EMERGENCY COMMANDERS

### Data & Operational Overload

Designed to support overwhelmed emergency commanders by providing auto-generated, highly structured 3-bullet SitReps updated dynamically every 30 minutes.



## 04 / MUNICIPALITIES & INSURERS

### Verifying Post-Event Incident Data

Supports municipal authorities with comprehensive data for post-event planning, and equips insurance providers with highly verified incident logs for claims.



# A Robust, Scalable Tech Stack

## Built for Spatial Intelligence

### CLIENT & MOBILE LAYER

React.js/Next.js, Tailwind CSS, Mapbox GL JS for web. Flutter or React Native for offline caching on mobile.

REACT / NEXT.JS

FLUTTER

MAPBOX GL JS

### API & BACKEND LAYER

Highly performant, modern request processing layer powered by Node.js or Python FastAPI using GraphQL APIs.

NODE.JS

FASTAPI

GRAPHQL

### DATABASE & INTELLIGENCE

PostgreSQL with PostGIS extension for spatial queries. Gemini API for multimodal, Hugging Face for local NLP.

POSTGIS

GEMINI API

HUGGING FACE

### INFRASTRUCTURE & HOSTING

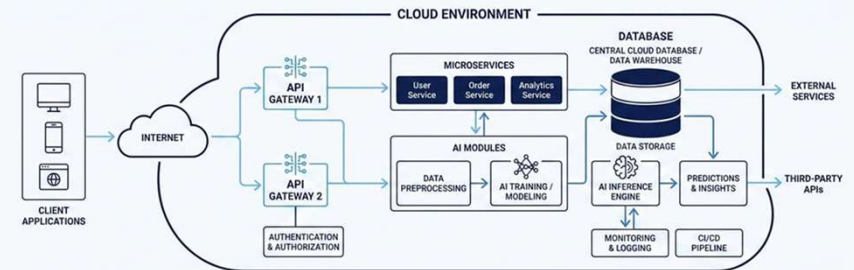
The entire containerized system is standardized via Docker, easily deployable to highly scalable GCP or AWS clouds.

DOCKER

GCP

AWS

### CLOUD PIPELINE ARCHITECTURE



Direct connection from clients over standard internet APIs into secure, containerized AI and microservices processing structures.

# A Deliverable Hackathon MVP

## With Clear Mitigation Strategies

### MVP Must-Have

- ✓ Online voice/image SOS submission
- ✓ GenAI extraction to structured JSON
- ✓ A GIS dashboard with heatmaps
- ✓ A triage scoring algorithm
- ✓ Manual or semi-automated volunteer dispatch

### MVP Should-Have

- Offline queuing
- Automated SitReps
- Skill-based matching

### Future Scale

- ◆ Horizontally scalable microservices
- ◆ Manage platform scale effectively

**Challenge: Dialects** Fine-tuning AI prompts for dialects

**Challenge: Connectivity** Using local PWA caching for low connectivity

**Challenge: Privacy** Implementing role-based access for victim data privacy

**Challenge: Platform Scale** Utilizing horizontally scalable microservices to manage platform scale effectively



# Faster Rescues, Equitable Prioritization, and Optimized Resource Allocation

## ↑ User Impact

# 01

Soteria delivers significant user impact by enabling safer, faster rescues without complex form-filling.

**KPI Tracked: SOS Submission to Volunteer Dispatch**

## ↑ Social Impact

# 02

Includes an equitable priority system that systematically weights vulnerable groups, such as the elderly and disabled, higher.

**KPI Tracked: Triage Scoring Accuracy vs. Actual Severity**

## ↑ Economic Impact

# 03

Optimizes the allocation of limited rescue resources, minimizing waste.

**KPI Tracked: Commander Time Saved via Automated SitReps**

## ↑ Operational Resilience

# 04

Secures data continuity during crisis situations with offline communication queues and resilient message recovery.

**KPI Tracked: Offline-Submitted SOS Processed Post-Reconnection**

## Platform Engine

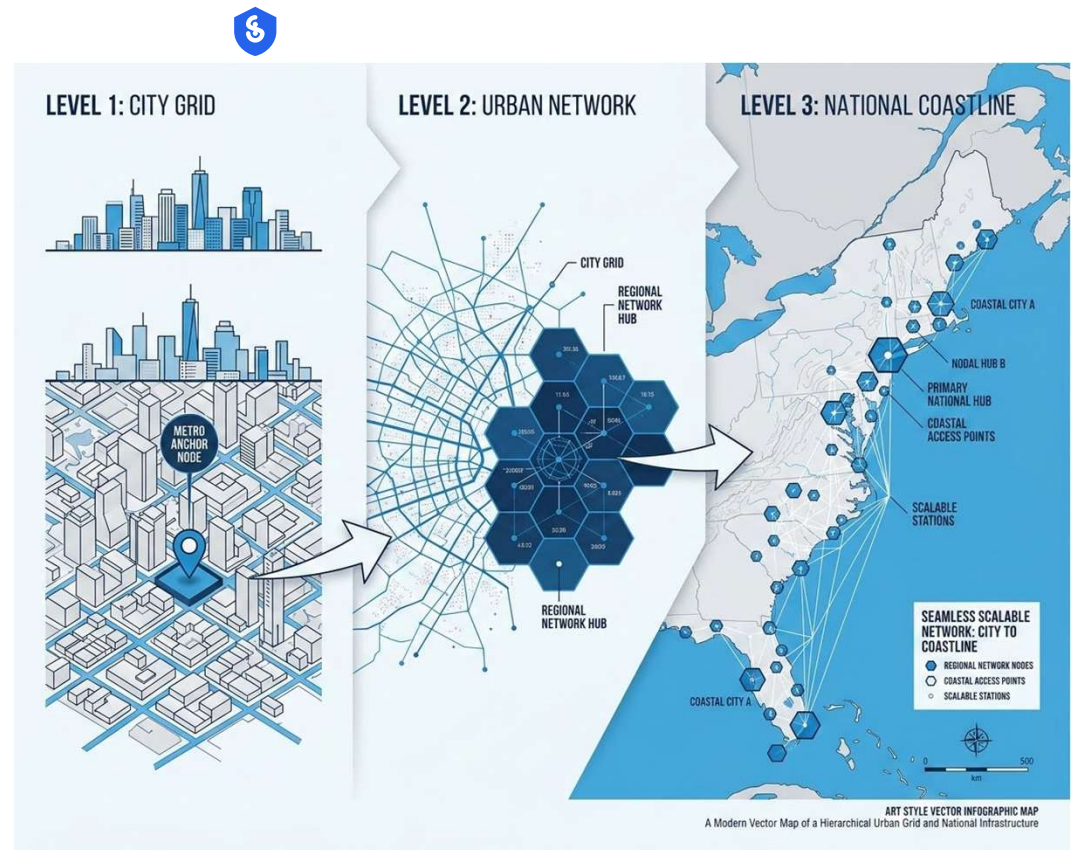
# A Disaster-Agnostic Engine Built for Global Scalability

### Technical Scalability

Soteria offers technical scalability through stateless, containerized backend microservices and PostGIS for optimized geo-queries.

### Sustainability & Monetization

Long-term sustainability and monetization will be driven by strategic licensing partnerships with State Disaster Management Authorities and NGOs, public safety innovation grants, and interfacing with insurance providers for secure, verified incident data.



### Geographic & Platform Scale

Geographic scalability is achieved via multilingual GenAI translation, enabling immediate deployment across different states and developing nations. The platform is disaster-type agnostic, easily scaling to earthquakes, cyclones, or industrial accidents.



## Deployment Strategy

# Staged Roadmap

Scale from MVP to Critical Infrastructure

### Phase 1: MVP

#### Core Triage & GIS

- Multimodal triage
- GIS Dashboard
- 0-100 triage scoring
- Basic citizen PWA dispatch

### Phase 2: Enhancements

#### Sync & Automation

- Offline-first syncing
- Automated SitReps
- Dynamic SOP briefings
- AI-verified task closures

### Phase 3: Scaled

#### Critical Infrastructure

- Predictive flood modeling
- Bluetooth mesh networks
- Multi-disaster templates
- IoT sensor & drone feeds



**SOTERIA**

# **A Future Where No Cry for Help is Lost in the Noise**

*"In a disaster, the difference between chaos and rescue is often just a few minutes—and a few minutes is exactly what Soteria gives back."*

Multilingual, unstructured data currently bottlenecks manual response efforts. Our multimodal GenAI solution structures and prioritizes voice and images in real time, delivering faster, fairer, and offline-capable closed-loop dispatches.

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**Thank you from Team Aflatoon:** Aryan Singh, Ayush Kumar Singh, Ayush Bhatt, and Abhijit Mukherjee